

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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CONTENTS

1. Introduction.....	3
1.1 Unit Processing Objectives	3
1.2 Use of document	3
2. Reference Documents	3
2.1 Company Standards.....	3
2.2 Codes and Regulations	4
3. Description of the Power Plant	4
3.1 Purpose of the Power Plant.....	4
3.2 Power Plant Configuration.....	4
3.3 Scope of Supply	4
3.4 Import Streams and Feeds	6
Specification of Ethane	7
3.5 Summary Process Description	7
4. Basis of Design.....	8
4.1 Design Objective	8
4.2 Design Case	8
4.2.1 Steam Cracking Unit	8
4.2.2 PDH Unit	9
4.3 Site and Climatic Data	9
4.4 Utilities	9
5. Configuration Study	10
5.1 Objectives and scope	10
5.2 Electrical studies – Generator stability	11
5.3 Reliability	11
6. Electrical Power System.....	11
6.1 Electrical Power Balance.....	11
6.1.1 Electrical Power Export Balance	11

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
--	---	--

6.1.2	Power Plant Internal Consumption	12
6.2	Electrical System and Grid Connections	12
6.2.1	Electrical Power Distribution Scheme	12
6.2.2	Power Plant Operation	12
➤	Turn Down Operation	13
6.3	Reactive Power	13
6.4	Electrical System Stability	13
7.	Steam AND Condensate Systems	14
7.1	Steam and Condensate	14
7.1.1	Alternative Steam Demand Scenarios	16
7.2	Steam and BFW Specifications	17
7.2.1	Pressures and temperatures	17
7.2.2	Steam Quality	17
7.3	Steam Condensate	18
8.	Specific Design Requirements	19
8.1	Unit Dependence	19
8.2	Black Start Capability	19
8.3	Plant Operating Lifetime	19
8.4	Sparing Philosophy	19
8.5	Availability	19
8.6	Turbine Fuels	19
8.7	Steam turbine	20
8.8	Heat Recovery Steam Generators	20
8.8.1	Auxiliary ("Fresh Air") Firing	20
8.8.2	Stack Exhaust Gas	20
8.9	Steam and Water Cycle	20
8.9.1	Deaerator	20
8.9.2	Condenser	21
8.9.3	Chemical Injection Packages	21
8.9.4	Boiler Feed Water Pumps	21
8.9.5	Blowdown Drums	21
8.9.6	Condensate Return System	21
8.10	Auxiliary boiler	22
8.11	Metering	22
8.12	Insulation and Winterizing	22
8.13	Layout	22
8.14	Schedule	22
8.15	Tools and Software	22
9.	Emissions and Wastes	22
9.1	Atmospheric Emissions	22
9.2	Waste Water Disposal and Water Minimization	23
10.	Interfaces and Tie-ins	23
11.	Battery Limit Conditions	23
12.	Process Guarantees	23

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
--	---	--

1. INTRODUCTION

1.1 Unit Processing Objectives

This document defines the design basis of the Power Plant to be constructed in the Integrated Petrochemical Complex (IPC) of Karabatan, Kazakhstan in order to provide electrical power to the IPC and its associated Infrastructures.

The Power Plant will also supply steam water to the whole Complex.

The Integrated Petrochemical Complex (IPC) will be constructed and commissioned in three distinct phases denoted "Phase 1", "Phase 2" and "Phase 3".

- Phase 1 (KPI) includes a Propane Dehydrogenation unit (PDH), a Polypropylene Unit (PP, 500 Kt/y) and associated utilities and offsite facilities.
- Phase 2 (KLPE) includes a Steam Cracking Unit (SCU), LLDPE/HDPE and LDPE units and associated utilities and offsite facilities. The Phase 2 IPC will produce an approximate 800 kt/y of polyethylene.
- Phase 3 (BD) includes Butadiene unit and associated utilities and offsite facilities to produce 250 kt/y of butadiene.

1.2 Use of document

This document shall be used as the basis for the Contractor to develop the Configuration Study and the front-end engineering design (FEED) package for the power plant supplying electrical power and steam to the IPC. The results and recommendations of the configuration study will be reviewed by Contractor and Company and a configuration selected, after which Contractor will develop the FEED on the basis of the selected configuration.

In the present document, the term "Contractor" refers to the FEED contractor of the Power Plant.

2. REFERENCE DOCUMENTS

The following document, pertaining to the engineering study of the Power Plant, should be used as reference applicable to the present project:

- Basic Engineering Design Data (BEDD) –
- Process Design Guidelines –
- Electrical Specification General–
- Instrument Specification General

2.1 Company Standards

Project Standards will be developed for Detail Engineering studies and as a minimum will comply with the requirements of Kazakhstan regulation.

The Project Standards are to be based on the following in descending order of priority:

1. Mandatory Kazakh standards
2. Licensors' requirements
3. International American/European standards

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
--	---	--

2.2 Codes and Regulations

The International American/European standards / codes applicable to the Project are in general:

- IEC, Euro Norms & ISA will apply for Electrical and Instrumentation
- ATEX, API, ASME and ANSI Codes will apply to Piping and Equipment
- ACI, AISC, ASCE and ASTM will apply to Civil and Structural.
- NACE, NFPA, ISO, FM, as applicable

3. DESCRIPTION OF THE POWER PLANT

3.1 Purpose of the Power Plant

The Karabatan Power Plant shall basically consist of a combined cycle power plant designed to generate and supply:

High voltage electrical power at nominal 110 kV is delivered to the Integrated Petrochemical Complex and to the associated infrastructures.

HP steam is supplied to various process and utility users in the IPC for start-up or shut down cases. Each IPC phase will generate MP and LP steam by their own lay down stations.

3.2 Power Plant Configuration

The Power Plant configuration comprises a set of identical gas turbo-generators, each with its associated HRSG, and a set of identical steam turbo-generators with condensers or a common condenser (to be confirmed by PWP Contractor). The HRSGs generate steam and feed steam turbines (STG) that supplement the electrical power generation. Exhaust steam from the steam turbines is condensed under vacuum by air-cooled condensers.

In preliminary analysis, the configuration comprises:

- Four (4) identical GTGs of ISO rating in the approximate range of 50 MWe each, with its associated HRSG;
- Two (2) identical condensing STGs of approximately 50 to 100 MWe and air-cooled exhaust condenser.
- One (1) package-type auxiliary boiler to provide low pressure steam for start-up.

PWP Contractor together with Vendors shall investigate the size of STGs in order to convert all the heat into electrical energy when the steam demand from IPC is minimized and to get a maximum efficiency of the power plant.

The Power Plant exports HP steam to the process units during some intermediate phases (PDH and SCU start-up or shutdown).

Beside of the above, steam will be required from power plant during the precommissioning for steam blowing and the like. These steam requirements will be confirmed later.

PWP Contractor shall optimise the HP steam generation, taking into account economic and operability criteria.

3.3 Scope of Supply

This section list the scope of supply of the PWP Contractor and it defines the battery limits. The PWP Contractor will be responsible for the following equipment:

1. GTGs and all associated equipment (lubrication system, fuel gas system, air inlet system, exhaust system, generator control and protection, etc).

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
--	---	--

2. HRSGs and all associated equipment (fans, stacks, dampers, fuel gas system, drains and blowdown system, chemical dosing, platforms, etc).
3. STGs and all associated equipment (lubrication system, condenser and condensate pumps, generator control and protection, etc).
4. Deaerator and feedwater pumps.
5. Generator breakers and generator transformers.
6. Auxiliary transformers.
7. HV switchgear and substations.
8. Cables/bus ducts from generators from/to: generator terminals / generator breakers / generator transformers / HV substation / auxiliary transformers
9. Low pressure steam supply for internal consumption (deaerator, heat tracing, anti-icing, etc).
10. Auxiliary boiler for start-up steam supply
11. Distributed Control System, rack rooms and control room.
12. DC power supply, battery rooms and UPS.
13. MV and LV substations supplying the PWP Contractor's equipment.
14. Grounding, earthing and lightning protection of the PWP Contractor's equipment.
15. F&G detection/fighting for the PWP Contractor's equipment.
16. Interconnecting piping and cabling between PWP Contractor's equipment.

The following equipment is excluded and will be supplied by Others. Where required the PWP Contractor will supply design and engineering inputs.

1. Demin water treatment, condensate treatment and waste water treatment facilities.

The battery limits of the PWP Contractor's scope of supply are as follows:

1. Electrical:
 - a. Secondary winding terminals of two 50 MW incoming transformers Output terminals of the HV feeder breakers
2. Mechanical:
 - a. Piping for fuel, import LP steam, condensate, demin water, export HP steam and other utilities will be supplied by Others up to the boundary fence of the power plant.
3. C&I and F&G:
 - a. Exchange signals from PP DCS to IPC Complex main control room.
 - b. Exchange signals between Power Plant DCS and Main Substations of all the phases (SCADA) –mainly for the load shedding
 - c. Output signals from the independent PP F&G system to IPC Complex F&G system.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB-4210-0001
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3.4 Import Streams and Feeds

During its operation, the Power Plant will receive the following import streams and feeds:

- Natural gas
- Ethane as back-up fuel gas
- Demineralised water from the water treatment plant
- Return Condensate from the condensate treatment plant
- Hot water from the hot water system (used in the HVAC system and for winterisation)
- Other utilities such as service water, potable water, compressed air, nitrogen, etc.

Natural gas is imported from the NCC (Makat - North Caucasus) pipeline into the fuel gas system, where it is metered and let down to an intermediate pressure of about 20 to 30 barg, or to another value (to be specified by PWP CONTRACTOR) compatible with its utilisation by gas turbines and other users. From there, the required flow rate is then transferred to the Power Plant. The natural gas is expected to have the following typical composition and properties [ref.: *Laboratory Akkol, Atyrau, KZ, analysis report No. 946 of 29.09.2011*]:

Parameter	GOST specification	Laboratory Measures Values
Heating value lowest Kcal/m ³	≥ 7600	8105
Wobbe index value range Kcal/m ³	9850 ÷ 13000	11705
Weight concentration of hydrogen sulfide, g/m ³	≤ 0.007	0.0014
Weight concentration of Mercaptan sulphur, g/m ³	≤ 0,016	0.0021
Volume ration of oxygen, %	≤ 1.0	0.0021
Solids particles,	≤ 0,001	Not present
Estimated composition of Natural Gas from Kaztransgas Pipeline		
Components,	% Vol	% Vol
CH ₄	94.609	94.573
C ₂ H ₆	2.569	2.584
C ₃ H ₈	0.473	0.480
i-C ₄ H ₁₀	0.076	0.078
n-C ₄ H ₁₀	0.101	0.105
i-C ₅ H ₁₂	0.020	0.021
n-C ₅ H ₁₂	0.022	0.023
ΣC ₆ H ₁₄	0.073	0.080
Nitrogen	1.376	1.364
Carbon Dioxide	0.684	0.686
Water Dew point, °C at T _{gas} =37.2°C & P _{gas} =50,7 kg/cm ² _g		-4.0
Water Dew point, at T _{gas} =_____ & at P _{gas} =40,0 kg/cm ² _g		-6.6
Density, kg/m ³ , at 20 °C and 760 millilitre of mercury (design)		0.710
Density, kg/m ³ , at 20 °C and 760 millilitre of mercury/pycnometer		0.712

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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Ethane: in the event that natural gas is not available for periods of time (say several days per year) ethane gas can be made available and the system is to be designed to accommodate this alternative fuel.

Specification of Ethane

Components,	% Vol
CH ₄	0.9795
C ₂ H ₆	98.9299
C ₃ H ₈	0.0858
n-C ₄ H ₁₀	0.0000
i-C ₄ H ₁₀	0.0000
n-C ₅ H ₁₂	0.0000
i-C ₅ H ₁₂	0.0000
Nitrogen	0.0000
Carbonyl sulfide(COS)	0.0004
M-Mercaptan	0.0000
E-Mercaptan	0.0000
Hydrogen Sulfide	0.0043
Carbon Dioxide	5 ppmv(10ppmw)

Demineralised water is produced in the water treatment plant in the IPC common utilities area and supplied to the power plant as a make-up to the steam and condensate cycle losses. Demineralised water is specified with an electrical conductivity below 0.2 μ S/cm at 25°C and a maximum SiO₂ content of 0.01 mg per litre. The demineralised water plant is excluded from the PWP Contractor's scope.

Steam condensate is returned from steam consumers to the condensate treatment plant in the IPC common utilities area for treatment. After treatment it is supplied to the power plant. Condensate is specified with an electrical conductivity below 0.2 μ S/cm at 25°C and a maximum SiO₂ content of 0.01 mg per litre. The condensate treatment plant is excluded from the PWP Contractor's scope

3.5 Summary Process Description

The process description hereafter is preliminary and for information only. PWP Contractor shall confirm and update as required the power plant's process description as part of the configuration study.

Electrical power is primarily produced by electrical generators driven by a set of gas turbines fired with natural gas from the IPC fuel gas system and steam turbines. The exported electricity is stepped up to 110 kV after the required electrical power has been levied for the operation of the power plant.

The exhaust gas of each turbine is directed through a dedicated heat recovery steam generator (HRSG) and used to generate primary steam at approximately 80 bar and 550 °C (*preliminary estimates*). An alternative design pressure is in the range 110 bar to 130 bar and the Contractor should select equipment from his standard range that optimises efficiency and capital cost. The exact pressure and temperature of primary steam from the HRSGs shall be determined and advised by PWP Contractor, as part of the configuration optimisation study. Primary steam is fed to a set of

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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condensing steam turbo-generators generating additional electrical power. The exhaust steam of each turbine is condensed in an air-cooled condenser, at the pressure allowed by ambient atmospheric conditions. Turbine condensate is picked up by pumps and returned to the deaerator, via the possible necessary treatment or polishing, as advised by PWP Contractor.

The HRSGs shall be fitted with post-firing systems to increase the steam output, if required to meet the needs of the steam turbo-generators and the specified export to IPC users. The firing system shall allow for auxiliary firing (i.e. with the gas turbine in service) and fresh-air firing (gas turbine out of service).

Alternatively, if economically justified, some of the steam might be generated by independent package steam boilers in order to delete or reduce the post-firing load.

If economically justified, low pressure steam will be supplied from a suitable extraction on the turbine. A pressure reduction station will serve as a 100% backup.

The deaerator use LP steam to strip the incoming mixture of demineralised water and recovered condensate and remove dissolved oxygen and carbon dioxide. Deaerated water is stored in the deaerator tank and picked up by the boiler feed water pumps. Conditioning chemicals such as oxygen scavenger and amine solution are dosed into the suction line of the boiler feed water pumps. BFW pumps supply boiler feed water to the power plant's HRSGs.

In order to supply steam to the deaerator, turbine glands, etc during start-up, an auxiliary boiler will be provided. This boiler will be a gas-fired, low pressure unit, supplying saturated steam. The PWP Contractor will select the capacity in order to satisfy the steam demand during start-up.

A chemical solution of phosphates, or equivalent, is injected into the steam drum of each HRSG in order to prevent fouling of the heat exchange surfaces. A continuous blow-down purge is bled from the HRSGs and flashed to recover some steam into the LP network or for the deaerator. The remaining liquid phase is flashed at atmospheric pressure, together with the intermittent blow-down purge. The resulting liquid effluent is discharged to the Waste Water Treatment unit of IPC.

4. BASIS OF DESIGN

4.1 Design Objective

The Power Plant will be constructed and started in two successive stages denoted "Phase 1" and "Phase 2". It is expected that the phases will be only a few months apart. Phase 1 will cater primarily for the power demand of the PP plant as well as the utility units. Completion of Phase 2 will provide for full power demand including that of the PE unit.

The design objective of the Power Plant is to provide for all the Phases the electric power required to meet the consumption of the IPC facilities in normal operation, start-up and any other operating situation and HP Steam for start-up and shut down situations.

The configuration of the power plant shall be able to handle the impact of the emergency shutdown of a single process unit on the utility balances and power distribution.

No simultaneous start-up of individual process units is assumed (cascade operation).

At each project Phase, the installed power plant capacity shall allow generating the required electrical power amount in any season within the range of climatic data specified in § 4.3.

4.2 Design Case

4.2.1 Steam Cracking Unit

Steam Cracking Unit will be designed according to Linde technology. During the start-up and shut down the unit will need to import an estimated 260 t/h of HP steam. About 80% of this amount will return to Power Plant as the condensate.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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4.2.2 PDH Unit

The PDH unit will be designed according to Lummus technology (Catofin).

During regular operation, all steam required within the PDH unit will be internally generated by waste heat boilers. During start-up and shut-down the unit will need to import an estimated 30 t/h of HP steam.

4.3 Site and Climatic Data

The site is located in the Republic of Kazakhstan by the northern coast of the Caspian Sea within 50 km of the city of Atyrau. The territory of the Atyrau region represents a low flat country or a slightly elevated flat country, located in a semi-desert and desert zone. Detailed climatic and site data are contained in the Basic Engineering Data (BEDD) Document No IPCI2-PGL-8110-DB-0000-0001

The Power Plant shall be able to generate steam and electric power to its design capacity, in the summer and winter conditions specified hereafter, as well as in any intermediate climatic condition.

Design summer conditions:

- Ambient temperature : 40 °C
- Atmospheric pressure : 992 mbar
- Relative humidity : 46 %

Design winter conditions:

- Ambient temperature : -20 °C (to be confirmed by PWP Contractor)
- Atmospheric pressure : 1023.5 mbar
- Relative humidity : 82 %

The temperature of -20 °C is the estimated value below which ambient temperature is expected to drop for no more than 5 % of the time during the four coldest months of the year. The specified absolute minimum ambient temperature, as per BEDD, is -38 °C. PWP Contractor shall allow for any possible impact of extreme low temperatures on the design and operation of the power plant.

4.4 Utilities

The utilities required for the operation of the power plant are provided by the IPC, including:

- Service water
- Demineralized water;
- Treated condensate
- Hot water
- Cooling water;
- Plant air;
- Instrument air;
- Nitrogen;
- Potable water;
- Fire-fighting water.

Due to the shortage of raw water at the site, PWP Contractor shall ensure that water use within the power plant is minimized. In particular, the condensing duty of steam turbines is achieved by air-cooled condensers.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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Utilities are delivered to the inlet battery limit of the Power Plant, at the conditions specified in the *Basic Engineering Design Data* (BEDD) document.

5. CONFIGURATION STUDY

5.1 Objectives and scope

The Power Plant will be designed and constructed as described in § 4.1, “Design Objective”.

For maintenance reasons, a set of identical gas turbo-generators (GTGs) and a set of identical steam turbo-generators (STGs) shall be selected and installed. PWP Contractor shall select the optimum design of HRSGs. PWP Contractor shall justify his choice on the basis of technical, operability and economic criteria.

The HRSG shall be provided with condensate preheater to recover heat from flue gas exhaust and to minimize the steam consumption for deaerator.

Optionally, PWP Contractor may include, in the configuration, steam generation by independent package steam boilers in order to reduce the HRSG post-firing load, if economically justified. Such configuration, if opted for, shall be proposed as an option in addition to the base configuration comprising post-firing or auxiliary firing of the HRSGs.

Configuration assessment:

The recommended configuration shall be selected to ensure optimal operability, flexibility and maintainability of the plant, as well as on the basis of economic criteria. PWP Contractor shall provide technical and economic justification of the selected configuration.

PWP Contractor shall assess the Total Installed Cost (CapEx) and Operating Costs (OpEx) for proposed configuration, highlighting the individual differential costs, advantages / disadvantages, impact on operability, etc., in order to enable Company to procure long-lead items of equipment (GTGs, HRSG and STGs).

PWP Contractor shall define the configuration, and develop the Configuration Study for the Power Plant based on the following criteria:

- The available grid as back-up resource is limited in capacity between 30 to 50 MW.
- Overall reliability: In order to provide a plant achieving the required reliability level (see § 8.5), PWP Contractor shall consider the appropriate level of equipment and driver sparing required.
- Layout space must be adequate to cater for future expansion without excessive capital cost (for example one extra gas turbine and one steam turbine).
- Selected critical machines should have a record of successful operation in Kazakhstan or surrounding region; supplier must have an established service infrastructure in Kazakhstan or have an association with an entity in Kazakhstan for such service.
- The critical machines, esp. GT's, ST's and generators, should be standardized, considering spares availability in the surrounding region and reducing maintenance time.
- The configuration of the power plant shall be able to handle the impact of the emergency shutdown of a single process unit on the utility balances and power distribution.
- No simultaneous start-up of individual process units is assumed (cascade operation).

The Configuration study will be submitted to Company for selection and approval before developing the design of the Power Plant.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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5.2 Electrical studies – Generator stability

PWP Contractor shall develop a system model using the ETAP design software for electrical power systems, or equivalent software tool, to investigate the electrical stability of the proposed system, taking account of disturbances within the IPC for the likes of start-up of high-power motors such as extruders, etc.

PWP Contractor will optimize the voltage levels within the Power Plant, combining generation levels, grid levels and transmission levels, in order to provide a rationalized electrical network.

Refer to major users list provided in Attachment 1. The design shall be such that starting currents of large drives listed do not impact the operation of other running motors within the IPC.

PWP Contractor shall include in the configuration study the following analyses:

- Transient analysis (bus fault, loss of generators);
- Short-circuit analysis;
- Load distribution analysis and optimisation.

5.3 Reliability

PWP Contractor shall evaluate the global inherent reliability of the Power Plant, defined as MTBF / (MTBF + MTTR), based on availability records of the single components (GTG and STG) involved in the design of the power plant.

PWP Contractor shall advise the predicted figures for availability and reliability of the Plant.

6. ELECTRICAL POWER SYSTEM

6.1 Electrical Power Balance

6.1.1 Electrical Power Export Balance

The required electrical power generation capacity of the Power Plant is based on the estimated demand from the IPC Complex. These demands, summarised hereafter, combine into an estimated net electrical power export balance from the power plant:

User	Power Demand	Start-up	Shut-down
Phase 1 (KPI)	55 MW	18-20 MW	8 MW
Phase 2 (KLPE)	102 MW	30-40 MW	10 MW
Phase 3 (BD)	34 MW	10-15 MW	8 MW
Water Treatment Plant	8 MW		
Air/Nitrogen Plant	8 MW		

Notes:

- (1) The normal power consumption is the expected active power (in MW) absorbed at the final consumer's electrical terminals during continuous steady operation at the unit's nameplate capacity.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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- (2) The maximum absorbed load corresponds to the highest steady power consumption anticipated in occasional or exceptional plant operating conditions, such as start-up, alternative operating mode, operation beyond design capacity, etc. Normal and maximum absorbed loads are obtained using design margins on available data.
- (3) Electrical power consumption of the polypropylene units includes the associated logistic facilities.
- (4) Electrical power loads of the units are preliminary estimates provided by the process technology licensors and shall be confirmed by PWP Contractor during the engineering design phase.
- (5) Non-process load includes such users as buildings, lighting and other infrastructure.
- (6) Power losses through the electrical distribution system include esp. cables and transformers. In preliminary approach, these losses are estimated at 10 % of the absorbed electrical power at users.

The above electrical power consumptions are preliminary and will be confirmed by the EPC Contractor during the engineering design phase.

The electrical losses across the distribution system are preliminary estimates. PWP Contractor shall advise his own estimate of distribution losses and take into account the impact on the design of the power plant.

6.1.2 Power Plant Internal Consumption

The internal electrical power consumption of the Power Plant must be added to the above export amounts and duly included in the overall power generation capacity of the Plant.

Presently, the regular internal electrical consumption of the power plant is estimated at about 7 MW and the design maximum consumption at 8 MW. PWP Contractor shall provide his own assessment of the power plant's internal electrical consumption and update this preliminary estimate as required.

6.2 **Electrical System and Grid Connections**

6.2.1 Electrical Power Distribution Scheme

The Power Plant switchyard will be connected by a bi-directional 110-kV overhead line to the public overhead grid of the Atyrau area. The connection of the overhead line to the public grid is protected by circuit breakers controlled by the grid operator.

For information, the capacity of the external 110 kV grid system is limited to around 50 MW.

6.2.2 Power Plant Operation

➤ Normal Operation

During normal operation, in all ambient conditions, the Power Plant is providing the process steam and electric power to the IPC Plants.

The External grid at 110 kV is connected to the Power Plant Substation with two symmetric lines each one of 50 MW's capacity.

During normal operation the External Grid shall contribute to Power Plant frequency control.

Gas turbines may modulate the loads with the steam turbine in order to achieve this balance. This condition is required to limit the possible trouble due to any of HV switch opening on the IPC Plants.

Power plant shall normally operate in parallel mode (and alternatively, in island mode) with power import from, or export to the grid based on the zero electrical balance. The power plant control system must be able to support both modes, with seamless automatic and manual switchover between modes. Eventual import or export would be arranged with the grid operator. Therefore, during normal operation the power plant must be fully capable of supplying the power requirement of the IPC with no import from the Grid.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
--	---	--

PWP Contractor shall evaluate, as an option, the benefit of importing and exporting electrical power from/to the Grid, and the impact thereof on the integrity and reliability of the Power Plant and of the IPC.

➤ Turn Down Operation

The Power Plant shall be able to follow the Process power consumptions, including steam, in the full range specified, as defined in the para 6.1.1 and 7.1 and in all ambient conditions.

The GTG turn down ratio in order to comply with emission limits which, at this stage based on the World Bank recommendation shall be 50% of Base Load.

➤ Abnormal Operation

The following scenarios, in all ambient conditions, shall be considered:

- a) During normal operation with Nr. 3 GTs running, in case of Nr. 1 GT trip: the remaining GT shall ramp up to the load set point available at the time before trip. If power output is not sufficient, the following steps shall be performed;
 1. external grid may integrate the missing power load requirement (instantly)
 2. load shedding shall switch of non-critical loads (instantly)
 3. the load shall be balanced by supplementary firing of HRSG in fresh air mode and increasing of the ST load.
- b) During normal operation with Nr 3 GT running, eg. during GT maintenance or in winter conditions, it is recommended to keep in operation in fresh air mode the HRSG correlated to the GT off, in order to supply additional steam to STG. In case of GT trip the HRSG correlated to the GT tripped shall ramp up in fresh air mode to the set point load available at the time before trip. If power output is not sufficient, external grid may integrate the missing power load requirement or the load shedding shall be activated. During this transition period the steam and power to IPC Plant shall not be influenced.
- c) In case plant is operating in conjunction with external grid to support the IPC Plant electric power consumption and Nr. 1 GT trip or the external grid switch open, it may be required to start a Load Shedding of the IPC Process Plant.
- d) In case a main unit / equipment is forced to shut down, the Power Plant margin must be considered during operation in order to cope with this scenario; sudden load reduction due to big user shut down from IPC Plant may be covered with GTGs in operation with external grid support or preventing this scenario keeping safe margin in loading dispatch. Calculation needs to be performed by PWP Contractor and GTG/STG Vendor

Power Plant may provide to IPC Plant the required HP steam for start-up using HRSG in operation with fresh air and with external grid in case GTGs are not available.

6.3 Reactive Power

In preliminary estimate, an indicative overall power factor of 0.85 is suggested for the electrical power users of the IPC.

PWP Contractor shall confirm this value and provide his estimate of the reactive power created in the different users, as well as in the transformation and distribution network, and include all required allowances in the electrical system design of the power plant.

6.4 Electrical System Stability

PWP Contractor shall investigate the stability of the electrical power generation and distribution system in case of:

- Start-up of large motors in the IPC;
- Sudden drop of electrical load due to unexpected shut-down of one or more consumers.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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The study shall consider the start-up of one major consumer at a time, based on the list of major electrical consumers included in Attachment 1.

The sudden load drop scenarios to be considered shall include, for Phase 1:

- Complete shut-down of any one process unit of Phase 1 and subsequent total drop of electrical load to that unit, while the rest of the IPC is running at its design capacity;
- Partial shut-down of any one process unit of Phase 1, causing a turndown to 60% of the normal power load, while the rest of the IPC is running at its design capacity;
- Shut-down of the entire polypropylene train, i.e. PDH unit and PP unit, causing a complete drop of the polypropylene train's maximum operating power load;
- Partial shut-down of the polypropylene train, causing a turndown to 60% of its normal power load;
- Accidental trip, and subsequent load drop, of any one extruder of the PP unit.

For Phase 2, the load drop scenarios to be considered shall include:

- Complete shut-down of any one process unit of Phase 2 and subsequent total drop of electrical load to that unit, while the rest of the IPC is running at its design capacity;
- Partial shut-down of any one process unit of Phase 2, causing a turndown to 60% of the normal power load, while the rest of the IPC is running at its design capacity;
- Shut-down of the entire polyethylene train, i.e. Steam Cracker, Butene-1 unit, LLDPE/HDPE unit and LDPE unit, causing a complete drop of the PE train's maximum operating power load;
- Partial shut-down of the polyethylene train, causing a turndown to 60% of its normal power load;
- Shut-down of the entire polypropylene train, i.e. PDH unit and PP unit, causing a complete drop of the polypropylene train's maximum operating power load;
- Partial shut-down of the polypropylene train, causing a turndown to 60 % of its normal power load;
- Accidental trip, and subsequent load drop, of the extruder of any one polymer unit.

For Phase 3, the load drop scenarios to be considered shall include

- Complete shut-down of any one process unit of Phase 3 and subsequent total drop of electrical load to that unit, while the rest of the IPC is running at its design capacity;
- Partial shut-down of any one process unit of Phase 3, causing a turndown to 60% of the normal power load, while the rest of the IPC is running at its design capacity;

7. STEAM AND CONDENSATE SYSTEMS

7.1 Steam and Condensate

The steam production capacity of the power plant must cover the full demand of the IPC in any operating situation.

The tables hereafter summarise, for all the Phases, the estimated steam consumptions of the IPC (without taking into account internal consumption of the power plant).

The balances are given for the expected normal operation, the maximum demand and the individual unit start-up situations (one unit starting up while the others run at normal capacity). The maximum

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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values correspond to the sum of individual units' maximum consumptions, and are not expected during the actual operation of the IPC. They constitute design values for the steam system.

				Total HP steam required for process (from power plant+utility boiler)	HP steam required for winterization			
					t/h			
KPI(phase 1)	import	winter	normal	0				
			start-up	0				
			shut-down	0				
		summer	normal	0				
			start-up	0				
			shut-down	0				
	export	winter	normal	1				
			start-up	20				
			shut-down	43				
		summer	normal	1				
			start-up	20				
			shut-down	43				
KLPE(phase 2)	import	winter	Normal (LP steam)	0				
			start-up(1)	0				
			start-up(2)	0				
			shut-down	0				
		summer	Normal (LP steam)	0				
			start-up(1)	0				
			start-up(2)	0				
			shut-down	0				
	export	winter	normal	118.4	166.9			
			start-up(1)	334.2 (To SCU : 261.4)	167			
			start-up(2)	350.2 (To SCU : 261.4)	167			
			shut-down					
		summer	normal	125.7	0			
			start-up(Note1)	334.2 (To SCU : 261.4)	0			
			start-up(Note2)	350.2 (To SCU : 261.4)	0			
			shut-down					
BD(phase 3)	import	winter	normal	0				
			start-up	20				
			shut-down	43				
		summer	normal	0				
			start-up	20				
			shut-down	43				
	export	winter	normal	0				
			start-up	0				
			shut-down	0				
		summer	normal	0				
			start-up	0				
			shut-down	0				
Polymer product	import	winter	normal	0				
			start-up	0				
			shut-down	0				
		summer	normal	0				
			start-up	0				
			shut-down	0				
	export	winter	normal	0				
			start-up	0				
			shut-down	0				
		summer	normal	0				
			start-up	0				
			shut-down	0				

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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The above values combine information submitted by process technology licensors as well as estimates for utility systems, offsites and infrastructures. All the indicated figures are preliminary and shall be confirmed to the PWP Contractor.

7.1.1 Alternative Steam Demand Scenarios

Beside the “normal”, “maximum” and “start-up” situations indicated in the above table, alternative operating situations of the Integrated Petrochemical Complex will include, in Phase 1:

- In normal operation of Phase 1 (KPI) steam demand is not considered
- PDH operating at normal capacity and PP unit shut down;
- PP operating at normal capacity and PDH unit shut down

In Phase 2:

- Normal operation of IPC at design throughput;
- Steam cracker shut down, all other units operating at normal capacity;
- Steam cracker starting up while all other units operate at normal capacity;
- PP train (PDH and PP units) operating at normal capacity, all other units shut down;
- Steam cracker starting up while LLDPE/HDPE and LDPE units are shut down, and PP train operates at normal capacity;
- PP train shut down, all other units operating at normal capacity;
- LLDPE/HDPE unit shut down, Steam Cracker at turndown, Butene-1 unit shut down;
- LDPE unit shut down, Steam Cracker at turndown, all other units operating at normal capacity.

All the above alternative operating scenarios should be considered in the power plant design. In the course of the engineering design phase, the PWP Contractor will confirm and identify the minimum, normal and maximum steam consumption flow rates, as well as possible alternative scenarios generated by the operation of the IPC and significant for the design of the power plant. The PWP Contractor will also establish and advise all continuous and intermittent consumptions of steam, including:

- Periodic or cyclic consumptions;
- Occasional consumptions, such as unit start-up, shut-down, maintenance, etc.;
- Accidental or emergency demands, i.e. occurring at unpredictable moments of the operation.

Based on an analysis of these continuous and intermittent consumptions of the IPC, PWP Contractor shall determine and justify the sizing of:

- The HP steam production capacity,
- The storage volumes of boiler feed water.

The overall system (steam and boiler feed water production + storage) shall be designed so as to cover the demand at all times during any operating phase of the installation's lifetime.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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7.2 Steam and BFW Specifications

7.2.1 Pressures and temperatures

Steam shall be produced as required and exported at the following conditions at the outlet battery limit of the Power Plant.

	Pressure, bar g			Temperature, °C		
	Min	Normal	Max	Min	Normal	Max
HP steam	42.7	43.8	tbc	395	404	tbc
VHP boiler feed water ¹	120	130	tbc	110	120	tbc

Note 1: As an option, VHP boiler feed water will be supplied to the SCU. The flowrate is 400 m³/hr and the PWP Contractor shall provide an optional design to satisfy this requirement.

These pressure and temperature values are preliminary and will be later confirmed or updated upon indications of the PWP Contractor, in the course of his engineering study.

The following mechanical design conditions shall apply to the networks of steam exported to the IPC:

	Mechanical design conditions	
	Pressure, bar g	Temperature, °C
HP steam	50.0 + full vac.	427
VHP boiler feed water	tbc	tbc

The above mechanical design pressures and temperatures are preliminary indications and shall be confirmed or updated by PWP Contractor during the present study. As much as practicable, piping pressure rating shall be kept in the most economical class.

The pressure and temperature conditions of Primary steam, used internally in the power plant to feed the STGs, shall be optimised by PWP Contractor for an optimal energy performance.

7.2.2 Steam Quality

The steam shall be of suitable quality for use in industrial steam turbines and process heat exchangers. In preliminary estimate, the following chemical characteristics are expected.

Total dissolved solids	mg / litre	0.5
Conductivity	µS / cm	< 0.2
Silica, as SiO ₂	mg / litre	< 0.02
Fe	mg / litre	< 0.02
Cu	mg / litre	< 0.003
Na + K	mg / litre	< 0.01
Oxygen	mg / litre	Nil

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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PWP Contractor shall confirm the chemical specification of the steam, based on applicable international standards.

Boiler feed water shall be produced inside the power plant by mechanical deaeration of demineralised water, followed by dosed injection of appropriate conditioning chemicals, including oxygen scavenger and amine solution. The chemical characteristics of boiler feed water shall be suitable for use in boilers and HRSGs operating at the different pressures:

- 80 bar, in preliminary estimate, for the HRSGs of the power plant, to be confirmed by PWP contractor.

In preliminary estimate, the following chemical characteristics are expected for deaerated and conditioned boiler feed water.

Regular BFW (50 to 80 bar g)

pH		> 8.4
Silica	mg / litre	0.01
Na	mg / litre	0.01
Fe	mg / litre	0.01
Cu	mg / litre	0.003
Dissolved oxygen	mg / litre	0.007
Organics	mg / litre	0.05

PWP Contractor shall confirm or update the characteristics of BFW to fit the quality requirements of the HRSGs.

7.3 Steam Condensate

Steam condensate will be supplied to the Power Plant from the condensate treatment plant at the following pressures and temperatures at the inlet battery limit of the Power Plant.

	Pressure, bar g			Temperature, °C		
	Min	Normal	Max	Min	Normal	Max
Clean condensate	3.0	3.5	4.5	tbc	100	110

The liquid pressures are understood at grade level. PWP Contractor shall confirm or update the required return pressure of condensate.

The following mechanical design conditions are expected for the return steam condensate systems:

	Mechanical design conditions	
	Pressure, bar g	Temperature, °C
Clean condensate	6.0	130

These values are preliminary and will be confirmed or updated by the PWP Contractor.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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8. SPECIFIC DESIGN REQUIREMENTS

8.1 Unit Dependence

The power plant will be capable to start and operate as an independent facility, yet subject to the availability of key utilities manufactured in the IPC. Since the power plant is required in advance of the IPC, the following utilities are deemed to be already available at the time of the power plant start-up:

- Demineralised water
- Cooling water
- Plant and instrument air
- Nitrogen
- Potable water
- Fire water

To pre-commission, start up and operate the above utilities necessary for the commissioning of the power plant, PWP Contractor shall foresee a connection to the public grid with the possibility of import/export of the power in amount of approximately 50 MW. PWP Contractor shall select the optimal option and justify his choice on the basis of operability, reliability and economic criteria.

8.2 Black Start Capability

N/A

8.3 Plant Operating Lifetime

Each Phase of the Power Plant shall be designed for an operating life of 25 years.

The Power Plant shall be able to continuously operate at its design capacity during at least 4 years between turnarounds for inspection and maintenance.

8.4 Sparring Philosophy

The number of gas turbines and steam turbines shall be sufficient to generate the required electrical power output during the summer season with one machine (GTG or STG) shut down. Similarly, the post-firing capacity of the heat recovery steam generators shall enable producing the adequate amount of steam to cover the export requirements as well as the steam turbo-generators' consumption in any season of the year, should any of four the gas turbines or HRSGs be unavailable.

8.5 Availability

For all of the Phases, PWP Contractor shall evaluate the global inherent availability of the Power Plant, defined as $MTBF / (MTBF + MTTR)$, based on availability records of the single components (GT and ST) involved in the design of the power plant.

PWP Contractor shall advise the predicted and guaranteed figures for availability and reliability and shall provide a list (and estimated cost) of any additional equipment that could be installed at Company's option to improve the availability and reliability of the Plant (2 out of 3 trip systems, or median select for controllers, for example).

8.6 Turbine Fuels

The regular fuel to the Power Plant and gas turbines shall be natural gas, supplied from the IPC fuel gas system.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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Additionally, gas turbines shall be designed to use Ethane as back-up fuel gas in case of loss of natural gas.

8.7 Steam turbine

The steam turbine will be of the condensing/extraction type. LP process steam (if required) will be supplied from suitable extraction points on the turbine.

The turbine will be equipped with a 100% bypass. The bypass will depressurise and desuperheat the steam before it enters the condenser. The bypass will be used during plant start-up and during failure of the steam turbine.

8.8 Heat Recovery Steam Generators

8.8.1 Auxiliary ("Fresh Air") Firing

PWP Contractor shall evaluate the option of providing auxiliary fresh air firing of the HRSGs, to provide continuity of steam supply to the STGs in the event of a failure of the GTGs.

Conversely, in case of failure of the STGs, the provision of a by-pass stack to enable the GTGs to operate in open cycle should be evaluated.

For operating situations requiring steam generation beyond the heat amount recoverable from the turbine flue gas and post-firing system, PWP Contractor shall evaluate the options of using fully fired HRSGs vs. additional stand-alone steam boilers. PWP Contractor shall issue a recommendation justified by operability, reliability and economic criteria.

8.8.2 Stack Exhaust Gas

Flue gas shall exit the HRSG stack at the lowest possible final temperature. However, a margin of 10 °C shall be maintained above the acid dew point corresponding to the flue gas composition.

Due to the expectable gas turbine firing temperatures, and considering the likely applicable regulations in Kazakhstan, PWP Contractor shall foresee the installation of a NOx abatement system in the exhaust gas duct of the heat recovery steam generators. In preliminary estimate, it is anticipated that such a system will operate with an injection of aqueous ammonia solution and contact with a catalyst bed on the flue gas path. The possible impact of the exhaust pressure drop generated by the catalyst bed shall be considered in the overall performance evaluation of the gas turbines.

8.9 Steam and Water Cycle

8.9.1 Deaerator

PWP Contractor shall size the deaerator for 120 % of the total demand for boiler feed water. PWP Contractor shall select the deaerator based on practical considerations such as the maximum vessel size, as well as the requirement for four years continuous running of the IPC between major plant turnarounds. Should PWP Contractor propose a single deaerator, he shall provide a technical and economic justification considering the capital cost and availability requirements. Each individual deaerator shall be designed for a possible turndown to 20 % of its normal production rate.

Mechanical deaeration shall achieve to the following specifications before any injection of conditioning chemicals.

- Oxygen content : 7 µg per litre
- CO2 : not detectable

Boiler feed water shall be stored in a tank in the base of the deaerator having a hold-up of at least 10 minutes of the maximum steady BFW production rate, between the normal and low liquid levels. The operating pressure and temperature of the deaerator shall be selected by PWP Contractor, taking into account possible requirements of Kazakh regulations. PWP Contractor shall justify his choice of

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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the deaerator operating conditions, based on operability and economic criteria. BFW must be supplied to the economisers of the HRSG at a temperature higher than the maximum predicted acid dew point of the HRSG flue gas in any operating situation. Depending on the deaerating temperature, PWP Contractor shall foresee BFW preheat facilities as required.

8.9.2 Condenser

The condenser shall be air-cooled and parts wetted by steam or condensate shall not be manufactured out of copper-containing material. Liquid ring vacuum pumps is the preferred solution for maintaining condenser vacuum.

8.9.3 Chemical Injection Packages

PWP Contractor shall provide the required chemical dosing and injection systems for adequate conditioning of the boiler feed water.

The specific chemicals to be injected shall be proposed by PWP Contractor. Phosphate injection would be envisaged locally at each steam generator, rather than in the BFW production unit.

The chemical injection packages shall be designed to treat, in continuous operation, 120 % of the maximum BFW demand of the complex. The storage capacity of the day tanks shall allow uninterrupted operation for at least seven days at the design BFW treatment capacity. The dosing pumps shall have adjustable flow rate, covering the range from 25 % to 250 % of the required injection flow rate at design BFW production capacity. The pumps shall be spared on an N+1 basis. Injection points shall be specified by PWP Contractor.

8.9.4 Boiler Feed Water Pumps

PWP Contractor shall develop the design of the Boiler Feed Water system depending on the final steam, condensate and boiler feed water balance. PWP Contractor shall optimise the selection of the BFW pressure level to be supplied from the deaerator and provide a justification for his selection based on the capital and operating cost.

A minimum of N+2 boiler feed water pumps shall be provided of boiler feed water. The maximum capacity of the BFW pumps and the hydraulic design flow rate of the network shall be at least 110 % of the expectable peak flow rate of boiler feed water to the installation, taking into account the likely coincidences of intermittent demands. In accordance with ASME requirements, the BFW pumps shall be able to deliver their rated flow capacity at a terminal pressure 3 % higher than the highest setting of any safety valve on the steam boiler.

The boiler feed water pumps shall be protected from discharge shut-off by a safe and reliable minimum flow recirculation device, such as a mechanical auto-recirculation valve or equivalent. Selected method shall be advised by PWP Contractor.

8.9.5 Blowdown Drums

A minimum of one continuous blowdown drum and one intermittent blowdown drum shall be provided for each boiler. The continuous blowdown system shall be sized for a minimum of 5 % of boiler Maximum Continuous Rating (MCR). The intermittent blowdown drum and system shall be sized for a minimum of 10 % of the boiler Maximum Continuous Rating (MCR). PWP Contractor shall provide a blowdown cooler if required.

The HRSGs' continuous blowdown shall be flashed to recover steam into the LP header or to feed the deaerator. The remaining liquid phase, together with the intermittent blowdown, shall be flashed at atmospheric pressure. PWP Contractor shall provide a cooling system to reject the blowdown stream at a suitable temperature for the downstream treatments.

8.9.6 Condensate Return System

Turbine condensate will be pumped from the condenser hotwell to the deaerator. In addition treated condensate will be supplied to the battery limits of the Power Plant.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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PWP Contractor shall advise the possible necessity of polishing the turbine condensate. The treatment plant will be supplied by Others.

8.10 Auxiliary boiler

The auxiliary boiler will be a gas-fired, low pressure unit, supplying saturated steam; typically a horizontal, fire-tube design. It will be a self-contained, skid mounted package, inclusive of all fans, burners, control systems and a stack.

8.11 Metering

Metering systems shall be installed at the power plant outlet battery limit, in order to measure and record the amounts of:

- Exported steam to each IPC phase
- Exported power
- Imported power for power plant internal consumption
- Imported condensate from the condensate treatment plant
- Imported demin water from the water treatment plant
- Imported fuel gas and liquid fuel
- All other imported utilities such as compressed air, nitrogen, service water, potable water, hot water etc.

8.12 Insulation and Winterizing

PWP Contractor shall include all the necessary winterizing, weatherizing and acclimatizing provisions in his design, including the selection of appropriate indoor/outdoor installation of equipment.

8.13 Layout

PWP Contractor is requested to define and provide the footprint plot area of the Power Plant

PWP Contractor shall ensure the adequacy of the allocated site plot of 420 m × 350 m, developed during an earlier study.

8.14 Schedule

The Power Plant is required to be on stream ahead of the RFSU of the IPC. For each Phase, the Plant should be able to meet the commissioning and start-up requirements of the IPC.

8.15 Tools and Software

PWP Contractor shall develop a model of the Power Plant using a specialised computer software such as GateCycle, GT Pro/GT Master or similar software.

9. EMISSIONS AND WASTES

9.1 Atmospheric Emissions

PWP Contractor shall advise and provide, an estimate of all atmospheric emissions expected from the Power Plant during its operation at nameplate capacity, in the design summer and design winter conditions, as well as in the ISO climatic conditions of gas turbines.

Emissions from the Power Plant shall be in accordance with the Environmental Design Basis, which will specify allowable limits for NO_x, SO_x and other pollutants.

BASIS OF DESIGN POWER PLANT KARABATAN	FOSTER WHEELER PROCESS ENGINEERING	1CD1439A-FWF-8110-DB- 4210-0001
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9.2 Waste Water Disposal and Water Minimization

Waste water will be collected and directed to the waste water treatment (WWT) plant located inside the common utilities area of IPC. Aqueous effluents shall be segregated into dedicated drainage systems, as specified in the reference document Waste Water Drainage Philosophy.

Possible liquid and solid wastes shall be identified by PWP Contractor

PWP Contractor shall provide, a tabulation containing full details and characteristics of all aqueous, liquid and solid waste streams from the Power Plant.

10. INTERFACES AND TIE-INS

PWP Contractor shall define all the interfaces and tie-in points between power plant and IPC Plants

11. BATTERY LIMIT CONDITIONS

All battery limit conditions shall be confirmed by PWP Contractor.

Battery limit conditions of returned steam condensate shall be confirmed or updated by PWP Contractor, during the design of the Power Plant's steam condensate treatment system.

12. PROCESS GUARANTEES

PWP Contractor shall advise the guaranteed parameters for the power plant performance of the Power Plant design. Guarantees shall include:

- Electrical power output, voltage and frequency.
- Electrical system stability in transient regimes.
- Chemical quality of steam.
- Chemical quality of boiler feed water.
- Quality of the treated return condensate, maximum purge rate and regeneration effluent.
- Steam and water losses of the power plant's internal steam cycle.
- Natural gas consumption at design steam and power production capacity, in summer and in winter seasons.
- Ethane consumption at design steam and power production capacity, in summer and in winter seasons, when gas turbines are fed with the alternative fuel.
- Electrical power consumption of the power plant.

Performance guarantees will be specified in detail in the relevant section of the contractual documents.